

Software Requirements Specification (SRS) Smart Campus Management System (Java Web Application)

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the Smart Campus Management System. It is intended for developers, project managers, teachers, and stakeholders to understand the system functionality and requirements.

1.2 Scope

The Smart Campus Management System is a web-based application developed using Java technologies. The system helps schools and colleges manage academic activities digitally through Student and Teacher panels.

1.3 Definitions

- Admin: System administrator managing the overall system
- Teacher: Faculty member managing subjects, attendance, marks, and projects
- Student: User accessing academic records and submissions

2. Overall Description

2.1 Product Perspective

The application is a web-based system accessible through browsers on desktop and mobile devices.

2.2 Product Functions

- User registration and login
- Student dashboard
- Teacher dashboard
- Attendance management
- Marks management
- Subject allocation
- Project submission system
- Project evaluation and marks allotment

2.3 User Classes

- Admin
- Teacher
- Student

2.4 Operating Environment

- Frontend: HTML, CSS, JavaScript
- Backend: Java (JSP/Servlets/Spring Boot)
- Database: MySQL
- Server: Apache Tomcat

3. Functional Requirements

3.1 User Registration

Students and teachers must register using email and password with role selection.

3.2 Login System

Secure authentication system with password recovery option.

3.3 Student Panel

- View attendance
- View marks
- Submit projects
- View project marks and feedback

3.4 Teacher Panel

- Manage subjects
- Mark attendance
- Add/edit marks
- Assign projects
- Evaluate projects

3.5 Attendance Management

Teachers update attendance and students can view attendance percentage.

3.6 Marks Management

Teachers enter marks and students can view subject-wise marks.

3.7 Project Submission System

Students upload project files and teachers evaluate them.

3.8 Admin Control

Admin manages users and monitors system activities.

4. Non-Functional Requirements

- Performance: System should load within 3 seconds.
- Security: Password encryption and role-based access.
- Usability: User-friendly interface.
- Scalability: Support multiple users.
- Reliability: Accurate attendance and marks records.

5. ER Diagram

Entities

- User(UserID, Name, Email, Password, Role)
- Student(StudentID, Name, Course, Semester)
- Teacher(TeacherID, Name, Department)
- Subject(SubjectID, SubjectName, TeacherID)
- Attendance(AttendanceID, StudentID, SubjectID, Status, Date)
- Marks(MarksID, StudentID, SubjectID, Marks)
- Project(ProjectID, Title, SubmissionDate, StudentID, TeacherID, ProjectMarks)

Relationships

- Teacher teaches Subject
- Student attends Subject
- Teacher assigns Project
- Student submits Project

6. DFD (Data Flow Diagram)

Level 0

Users interact with Smart Campus Management System.

Level 1 Processes

1. User Management
2. Attendance Management
3. Marks Management
4. Project Management

Data Stores

- User Database
- Attendance Database
- Marks Database
- Project Database

7. Use Case Diagram

Actors

- Student
- Teacher
- Admin

Use Cases

Student: Register/Login, View Attendance, View Marks, Submit Project

Teacher: Manage Subjects, Add Attendance, Add Marks, Assign Projects

Admin: Manage Users, Monitor System

8. Use Case Descriptions

Use Case: Submit Project

1. Login to system
2. Open project section
3. Select assigned project
4. Upload project file
5. Submit project

Use Case: Add Attendance

1. Login to system
2. Select subject
3. Select class/students
4. Mark attendance
5. Save attendance record

9. Conclusion

This SRS document outlines the requirements for the Smart Campus Management System. The system provides digital management of attendance, marks, subjects, and project submissions for schools and colleges.